A Nose-Teeth-Based System For Users With Restricted Upper Limb Mobility to Interact With Computer Devices

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***Abstract*—Multimodal interaction is a recent development in the field of Human-Computer Interaction that offers a variety of ways to interact with devices. The use of this technology allows those who are physically disabled to utilize technology effectively and work alongside physically fit individuals. Due to their physical limitations, people with physical disabilities are always one step behind society as a whole, which causes them to have inferior complexity and prevents them from using their hidden talents. This research aims to develop a platform for people with restricted upper limb mobility so they can use their noses and teeth to interact with their computer devices. With the help of the proposed system, the disabled can use their devices without assistance, and they can perform their daily activities which rely on computer devices. Many systems that use multimodal interaction for clinically disabled individuals have been developed with eye gaze, speech-based, and body gesture interaction. In contrast to the nose-teeth-based interaction, however, each of these modes of interaction has certain drawbacks. By studying the mode of nose-tooth interaction, this research seeks to develop a novel system. Apart from creating a system for the targeted user group, an assessment of the system's effectiveness, efficiency, and satisfiability is carried out through usability testing. Following the evaluation, approximately 87% of users report feeling at ease with the system, and 93% report having a decreased dependence on their using the computer device. Therefore, it can be concluded that computer users with limited upper limb mobility will greatly benefit from the proposed system. In summary, this study offers a nose-teeth interaction system that will enable people with limited upper limb mobility to participate significantly in society.**

***Index Terms*—Human-Computer Interaction, physically disabled, nose-teeth interaction**

I. Introduction

The term "upper limb" refers to the upward region of the body, which comprises the arms, shoulders, wrists, and hands. Limitation of movement in the upper limbs is referred to as upper limb restriction. There are several reasons why an upper limb may be restricted, such as congenital issues (defects from birth or genetic disorders), trauma, illnesses (arthritis, neurological disorders, muscle disorders, etc.), injuries from repetitive motions, improper medical interventions, vascular disorders, etc. Since the patient is unable to control their upper limb, upper limb restriction can have a major impact on their life.

The limitations can have a major impact on the patient's day-to-day activities, making it harder to do everything from simple self-care to difficult work-related tasks. People who are restricted in their upper limbs may find it difficult to remain independent and perform tasks that call for fine motor skills. Beyond physical restrictions, the effects also affect social interactions, emotional health, and technology use. The help they receive from others defines their lives. Being unable to use a mouse, keyboard, or touchscreen can be greatly impacted by restrictions in arm, shoulder, or hand mobility. It becomes more difficult to complete tasks that require typing, clicking, or precise cursor movements. This restriction may apply to technological use for entertainment, professional purposes, or even academic activities.

Upper limb restriction is a highly prevalent type of clinical disability. The World Health Organization estimates that 1.3 billion people worldwide are significantly disabled [1]. This equals one in six people on the planet or 16% of the total population. According to another study by the Pew Research Center, 67% of Americans with disabilities between the ages of 18 and 64 own a desktop or laptop, compared to 84% of people without disabilities in the same age range [2]. It makes it very evident that individuals with clinical disabilities tend to stay away from computers because they require assistance from others to operate them.

In the field of human-computer interaction, multimodal interaction is a relatively new development that provides multiple ways to interact with devices. Utilizing various modes of interaction to communicate with devices other than the conventional one is known as multimodal interaction. The five senses of the human body—sight, smell, hearing, taste, and touch—are matched with the modes. However, the current multimodal systems do not yet use taste or smell.

As the world grows more and more technologically advanced, new methods for interacting with computer devices have been developed in addition to the common practice of implementing the touch of fingers. There are many different

ways that multimodal interaction can occur, including body language and gestures, eye gaze, speech-based interaction, nose-teeth interaction, and more. The major purpose of these interactions is to assist people who are unable to use their gadgets conventionally, such as physically fit people.

There is a good deal of research on creating multimodal systems for people who are clinically impaired, but not much of it takes into account the circumstances of patients who have limited upper limb mobility. Eye-gaze-based interaction is widely utilized to help impaired people develop their ability to engage with technologies. Nevertheless, because the eye-gaze interaction's interaction area is so tiny, mistakes like picking an undesirable spot or acting inappropriately on the screen might occur with only a slight deviation in look. Another problem that could occur is the blinking of the eyes. Blinking can lead to several mistakes. A multimodal system can be developed using speech-based interaction, however, patients with hearing loss or who are dumb or inaudible cannot use it. Another method of engaging with systems is through body language and gestures, which people with upper-limb limitations are unable to employ.

We can rely on nose-teeth-based interaction to create a hand-free interaction system for people with limited upper limb mobility while taking into account all the disadvantages of other modalities of interaction.

With this research, our main objectives are to accomplish the following. Firstly, to create an effective nose-teeth-based interaction system that enables users with restricted upper limb mobility to communicate with computer devices. Secondly, to evaluate the developed system by employing usability testing to make a self-assessment.

This study's contribution is that the existing nose-teeth-based interaction system was heavily influenced by light. The systems would be less effective at night or in low light conditions. However, based on the evaluation results, it can be concluded that the proposed system works equally well at night or in low light.

The remaining chapters are organized as such: The related works have been discussed in Section II. The methodologies were discussed for the existing multimodal interaction system and the nose-teeth-based interaction system. Section III is a detailed description of the methodology of the system. The details about the development of the system carried out in this research are also presented in Section III. At first, the nose was tracked for cursor control. Then, the lips were detected for the click operations. Thus, the main purpose of this whole chapter is to give a brief discussion about the overall design of the system, including the development of the system. The process of system evaluation has been discussed in Section IV. Section V includes the result analysis of the evaluation of the proposed system. The ending chapter states the limitations and future works of the thesis.

II. Literature Review

There has been a significant amount of research done in this area for disabled people using computer devices to provide an alternative way to interact, so they face fewer obstacles and can have a significant impact on their daily lives.

M. Mangaiyarkarasi and A. Geetha [3] created a system that detects faces using Viola-Jones and Haar-like features to detect eye regions and track eye movements. The SVM method was also employed in the tacking of the eye for cursor control, as seen in [4]. P. Gyawal et al. [5] developed another system that uses eye tracking for face detection and is accompanied by the eye variance feature (EVF) and Gaussian mixture model (GMM) to track the eyes, and then uses the Finite State Machine (FSM) to detect the blinking pattern. Yan Shi et al. [6] used the CNN model for face detection and an end-to-end model for face expression recognition for cursor control in another system. Likewise, Rahib H. Abiyev and Murat Arslan used eye tracking for wheelchair control.

Apart from eye tracking, research on nose tracking for cursor control has also been conducted. Shadman Sakib Khan et al. [7] developed a system that uses the Viola-Jones algorithm to track the mouse using the nose point, and the nose is tracked by calculating the middle point for cursor control and smile detection for click. A similar method was used by Rohit Lal and Shital Chiddarwar [8] for face detection and nose tracking, as well as Haar-like features for eye tracking, with the exception that the smile is used for scrolling and left or right eye blinking is used for clicking. S. Kasun Chathuranga et al. [9] built another system that combined nose tracking with voice commands. Mireya Zapata et al. [10] used OpenCV and dlib to perform nose tracking as well as smile and voice commands.

Overall, several issues with the existing research were discovered. For instance, much research has been conducted on eye-gaze-based tracking, which requires a high level of precision that may be difficult for disabled people to achieve. Furthermore, eye detection is easily influenced by lighting conditions, eye shapes, eye sizes, and eye fatigue, making it difficult to use. Second, while the number of studies using the nose for cursor control is limited, it does necessitate the use of additional hardware devices. In some cases, the nose is assumed to be in the center and is not properly detected because it might vary. To address the research gaps and problems, we set the goal of developing a nose-teeth-based interaction system for computers, that does not require any hardware devices and detects the nose point accurately.

III. System Development

The following subsections provide a description of our system's development process.

A. Nose Tracking Cursor Control

We have used MediaPipe's FaceMesh for robust face detection and key facial landmark extraction on the live webcam input. Then we detected five nose points, including the nose tip, and the other points around the nose. We have calculated the distances and directions of the points around the nose from the nose tip and mapped them for cursor control as described in Fig.1

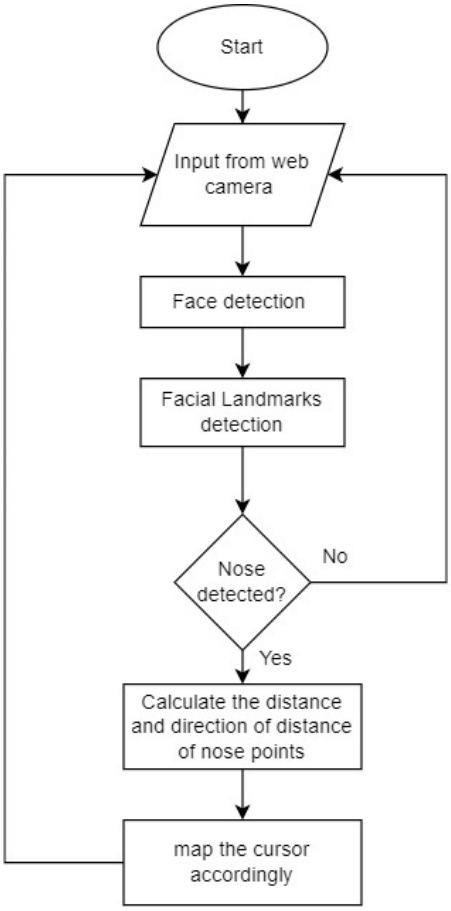


Fig. 1. Cursor Control Using Nose Tracking.

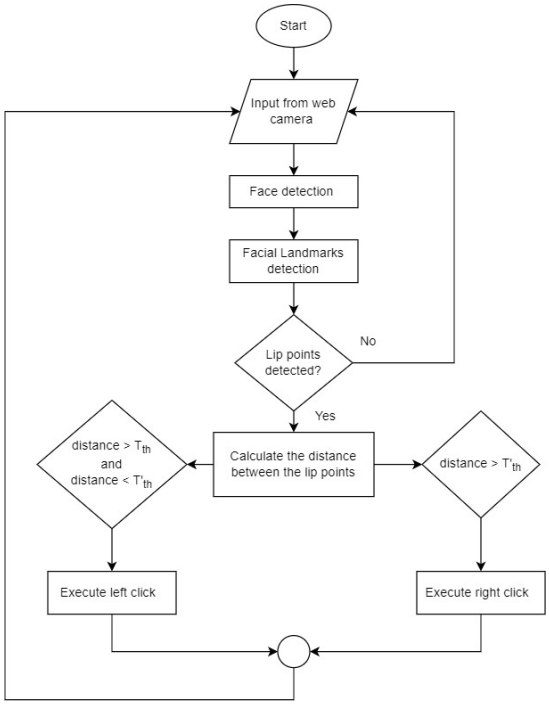


Fig. 2. Lip detection for clicking.

B. Lips detection with different operations

The MediaPipe's FaceMesh model was also used to detect the middle lowest point of the upper lip and the middle highest point of the lower lip, which are generally in the same place, and if the lips are unsealed, the difference between them increases, which is calculated for the click operation as mentioned in fig. 2. For opening the mouth, the click operation is done, and for opening the mouth wider, a right-click operation is executed. Two threshold values are considered for this T_{th}, T'_{th} for left click and right click, respectively, and they denote the distance between the lip points where $T_{\text{th}} < T'_{\text{th}}$. For operations like scrolling, double-clicking, and drag and drop, the respective mode has to be activated based on the mouse position on the screen, as mentioned in table I

TABLE I
DIFFERENT OPERATIONS AND THEIR ACTIVATING CONDITIONS

Conditions	Mode or Operation	Operation Description
Cursor is in the first half of the left corner of the display	Scroll	Scroll to the extreme up.
Cursor is in the second half of the left corner of the display	Scroll	Scroll to the extreme down.
Cursor is in the first half of the top corner of the display	Mode 0	Double clicks are executed.
Cursor is in the second half of the top corner of the display	Mode 1	Drag and drop operation is executed.

C. Automatic extended battery life

For extended battery life and controlled power consumption, the screen brightness is lessened whenever the face is not detected for some time, and upon detecting the face again, the brightness is set back to default as described in Fig. 3

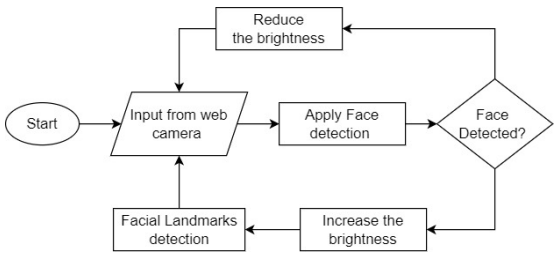


Fig. 3. Brightness Control for Extended Battery Life.

IV. EVALUATION

A. Initial Setup

To assess computer accessibility, usability testing was conducted with individuals with clinical disabilities. We used five different desktop devices and selected three tasks that represent common desktop activities. The following tasks were selected:

- Task 1: Open a PDF file and navigate it using extreme up-and-down scrolling.
- Task 2: Create a new folder.
- Task 3: Organize 3 files on a desktop (alphabetically).

These task scenarios were chosen to reflect common user challenges.

B. Participants Profile and Study Procedure

A total of 15 participants were invited to conduct the test. None of them were involved in the development process, and each of them was familiar with using computers. To explore the effectiveness of our developed system, three trials were conducted for each task, and a within-group study was conducted. For each of these tasks, three different metrics are recorded. The matrices are as follows:

- Efficiency.
- Effectiveness.
- Satisfiability.

For each of the metrics, several parameters were considered. For efficiency, the following parameters were calculated:

1) **Standard Task Completion Time(A):** The time taken for the user to complete the task. in
2) **Task Completion Time By The Evaluator(B):** The time taken by the evaluator or researcher to complete the same task as the user this Helps assess the task complexity and difficulty level from the evaluator's perspective.
3) **Time Deviation (B-A):** The difference between the task completion time by the evaluator (B) and the standard task completion time (A). This indicates the degree of deviation or discrepancy in task performance between users and evaluators and provides insights into user efficiency and potential areas for improvement.
4) **Number of clicks:**The number of clicks performed by the user to complete the task.
5) **Optimal click:**The minimum number of clicks required to accomplish a task efficiently this represents the ideal interaction scenario where users can achieve task objectives with minimal effort.

For effectiveness, the following parameters were calculated:

1) **Number of times asked for help:** Indicates how frequently users need assistance or guidance to complete tasks. A higher number of help requests may indicate difficulty understanding or navigating the interface. This identifies areas of the interface or tasks that users may find unclear or difficult to understand.
2) **Number of attempts/mistakes:** Each attempt or mistake made by the user is recorded to determine the task's difficulty or complexity.
3) **Success rate:** The percentage of successful task completions compared to the total number of attempts. Calculated as the inverse of the number of attempts, it provides a percentage of successful task completions and insight into users' overall effectiveness in completing tasks within the interface or application.

Lastly, for satisfiability post questionnaire was used.

V. RESULT ANALYSIS

The summary table of the evaluation is provided here for each of the parameters.

A. Efficiency

Each of the parameters is discussed below:

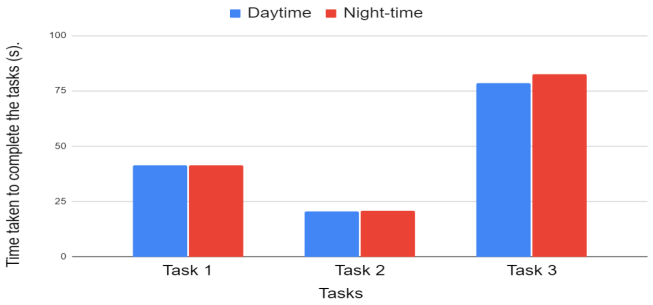


Fig. 4. Time taken (avg) for each of the tasks

1) Time taken to complete the task: : Here in Fig. 4 task 3 takes the most time followed by task 1 then task 2.

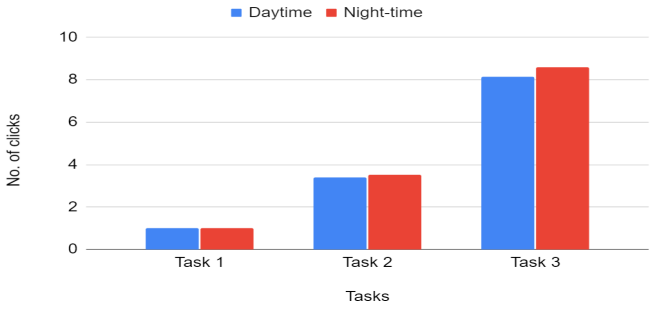


Fig. 5. No. of clicks (avg) for each of the tasks

2) No. of clicks:: Here in Fig. 5 task 3 requires the most number of clicks followed by task 2 and task 1. It can be seen that the number of clicks for the first and the second tasks is the same compared to the usual case.

3) Time Deviation:: This is the difference between the task completion time by the evaluator (B) and the standard task completion time (A). This indicates the degree of deviation or discrepancy in task performance between users and evaluators. Here in Fig. 6 task 3 has the most time deviation followed by task 2 and task 1. It can be seen that the number of clicks for the first and the second tasks is the same compared to the usual case.

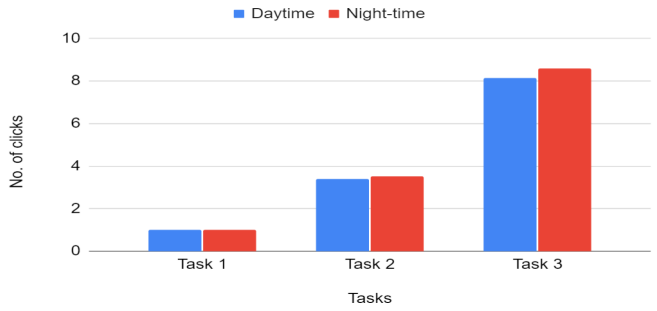


Fig. 6. Time Deviation (avg) for each of the tasks

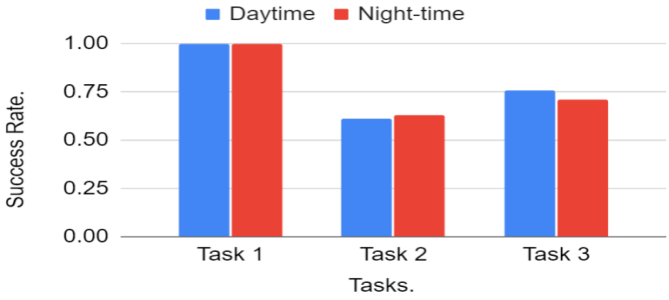


Fig. 9. Success Rate (avg) for each of the tasks

B. *Effectiveness*

Additionally, each of the parameters is discussed below:

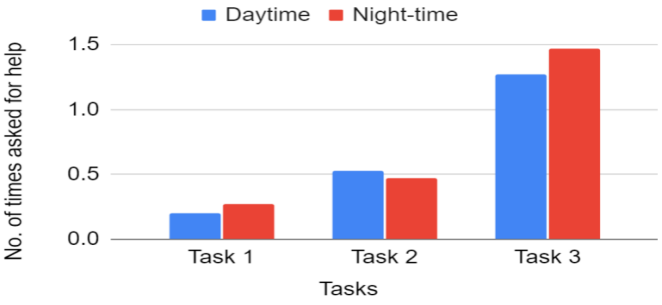


Fig. 7. No. of times asked for help (avg) for each of the tasks

1) No. of times asked for help: : Here in Fig. 7 task 3 requires most help followed by task 2 and task 1.

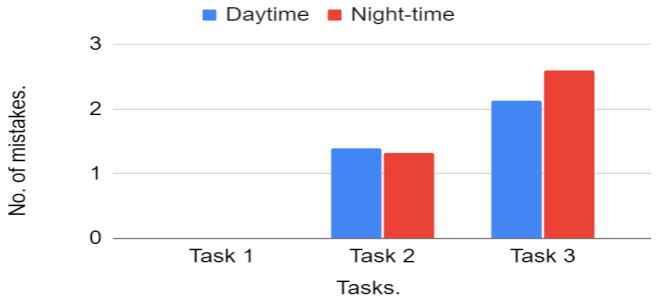


Fig. 8. No. of mistakes (avg) for each of the tasks

2) No. of mistakes: : Here in Fig. 8 task 1 includes no error and task 3 has the most error followed by task2.

3) Success Rate: : Here in Fig. 9 task 1 includes a 100% success rate while task 2 includes the lowest rate of success followed by task 3.

Overall we can see that for each of the parameters, the values for daytime and nighttime are very similar as for most of the other cases they do not work well at nighttime with only screen light and the values for the first two tasks are also adequate.

C. *Satisfaction*

A post-satisfaction questionnaire was used to measure the efficiency. The outcome of the questionnaire is given below:

1) For a rating out of 5, the average rating is 4 in terms of whether the system enhances comfort, satisfaction, and preference.
2) Around 87% of people feel comfortable with our system.
3) 93% of them state it increases their independence in terms of using the computer.

VI. Future works and conclusion

Future works may include mobile device compatibility, and increasing user accessibility across various platforms. Furthermore, addressing the impact of continuous camera usage on battery life is a top priority for future development. Improvements in click detection accuracy, particularly under varying conditions, are essential for improving user experience. Future research efforts may also concentrate on improving the system for more complex operations and investigating ways to integrate the technology into smartphones to increase user independence and comfort.

Overall, the nose-teeth interaction system has the potential to improve the quality of life for individuals with hand disabilities. It enables them to use digital devices more effortlessly, eliminating all barriers and allowing them to be more independent. With further testing and refinement, this system could become a valuable tool for enhancing the accessibility and inclusivity of digital devices for individuals with disabilities.

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